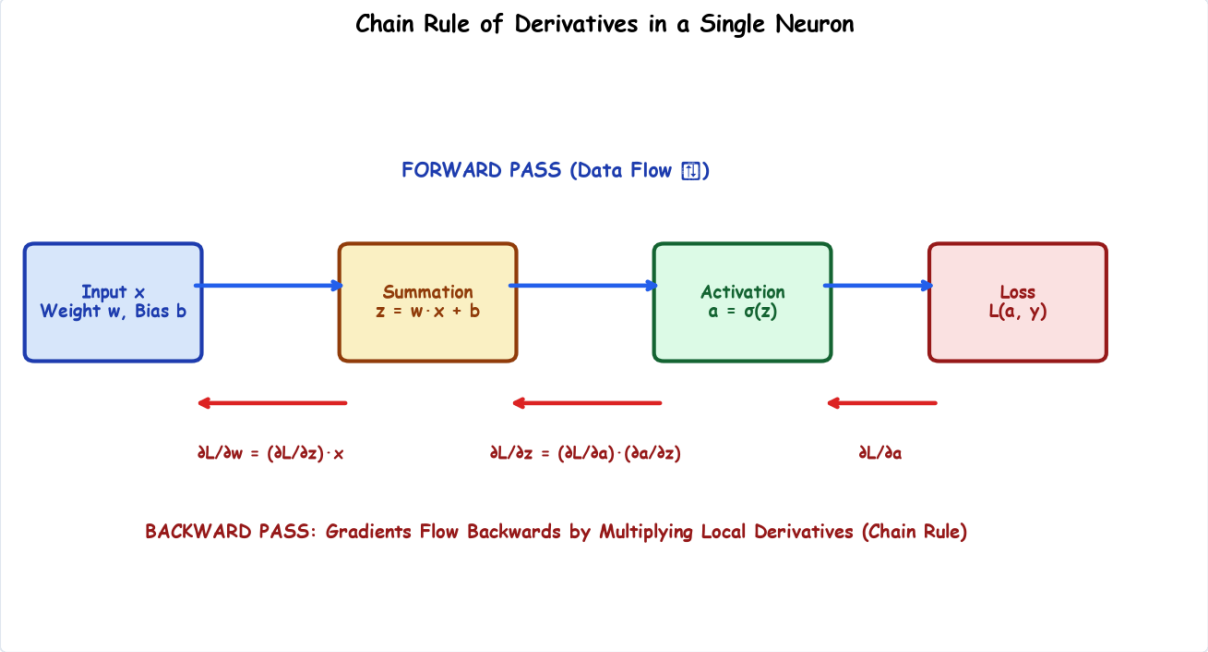
Submodule 37.05: Chain Rule of Derivatives in Neural Networks (Hinglish)

1. Chain Rule kya hai aur iski zaroorat kyun hai?

Calculus me Chain Rule ka seedha matlab hota hai: agar y depend karta hai u par, aur u depend karta hai x par, toh y ka derivative x ke respect me nikalne ke liye hum dono intermediate derivatives ko aapas me multiply karte hain:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$



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2. Neural Networks me Chain Rule ki Zaroorat

Neural network me jab hum loss nikaalte hain, toh loss formula ke andar direct weight w nahi hota!

- Loss $\mathcal{L}$ depend karta hai output prediction a par.
- Output a depend karta hai linear sum z par.
- Linear sum $z = wx + b$ depend karta hai weight w aur bias b par.

Dependency chain aisi banti hai:

$$w \longrightarrow z \longrightarrow a \longrightarrow \mathcal{L}$$

Isiliye agar hume weight update karne ke liye $\frac{\partial \mathcal{L}}{\partial w}$ nikaalna hai, toh hume Chain Rule lagana padta hai:

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

Har layer ka local derivative peeche se aane wale error signal ke saath multiply hokar sabhi weights tak pahunch jata hai!\n